

Adaptive Lexical Inference Under Input Uncertainty

Abstract

Human language comprehension operates under input uncertainty: readers resolve morphologically deviant but interpretable word forms either via morphological decomposition or Bayesian probabilistic inference. This behavioural study used neologisms to dissociate these competing cognitive architectures. Mandarin speakers completed self-paced reading and offline acceptability judgement tasks. Online reading revealed equivalent processing costs across neologism types, consistent with noisy channel Bayesian prediction. Yet, offline reflective evaluation showed mild preference for clipping-derived novel words. This online-offline dissociation indicates that lexical inference is not monolithic. Under resource-rational cognitive constraints, the brain dynamically re-weights inference strategies across real-time versus reflective processing regimes, reconciling structural analysis and Bayesian decoding.

Keywords: morphological processing, lexical inference, input uncertainty; noisy-channel Bayesian decoding

1. Introduction

When readers encounter a word form that violates canonical morphological conventions yet remains semantically interpretable, how does the native lexical processing system resolve the tension between surface form anomaly and recoverable meaning? Two broad families of accounts offer fundamentally different answers to this question. One class of theories posits that comprehension relies on structured morphological analysis: the processor decomposes complex forms into morphemic building blocks (Caramazza et al., 1988; Clahsen, 1999), and processing difficulty scales with the severity of morphological rule violation. An alternative framework conceptualizes language understanding as Bayesian inference over noisy input: surface deviations are treated as signal corruption, and readers prioritize recovering the most plausible intended meaning, potentially de-emphasizing fine-grained morphological distinctions during real-time interpretation.

Distinguishing between these competing perspectives has proven empirically challenging, because naturally occurring canonical linguistic input seldom creates controlled, systematic variation in morphological deviance while holding semantic interpretability constant.

Empirically testing these competing predictions requires well-controlled stimuli: forms that vary systematically in the structural nature of their morphological violation, yet remain interpretable in context. Naturally occurring novel words seldom satisfy this constraint, as real world lexical innovation tends to be idiosyncratic and hard to experimentally balance across confounds such as frequency, semantic plausibility, and contextual fit. The present study leverages a convenient source of such controlled material: systematically deviant word forms produced by second language (L2) learners of Mandarin Chinese.

Corpus research has documented these anomalous forms and built taxonomies to describe their structural properties (Xing, 2003). Early corpus-based work established multi-part taxonomies of learner-produced lexical deviations, documenting morphological over-generalisation that yields novel compound forms (Xing, 2003; Xu, 2004). Subsequent scholarship further refined these classification schemes: Zhang (2007) proposed a tripartite framework for L2 lexical deviations, while Gao (2008) expanded error typologies across semantic, collocational, and pragmatic dimensions, without addressing the underlying psycholinguistic representations that drive learner lexical innovation. Building on this corpus tradition, Wang (2018) carried out a key investigation of learner-produced novel word forms (named as neologisms), filtering confounds from phonology and orthography and deriving a six-category classification (see **Table 1**).

Table 1. Neological types with sample lexical items

morphological substitution	*内	边	VS	里	边	内 substitutes 里
	nèi	biān		lǐ	biān	
	inner	side		in	side	
						inside
blended misuse between word & morpheme	*花	公	园	VS	花	园 花园 as a whole word
	huā	gōng	yuán		huā	yuán can't be split.
	flower		garden		flower	garden

	garden							
forming a non-existent word by clipping	*古	楼	VS	古	代	楼	阁	non-existent clipping
	gǔ	lóu		gǔ	dài	lóu	gé	
	ancient	building		ancient		building	tower	
	ancient building							
existent structure but non-existent word	*比	赛	会	VS	比	赛	Only	运动会 (sports
	bǐ	sài	huì		bǐ	sài	meet)	exists.
	compare	match	meeting		compare	match		
	match							
substitution of relevant meaning	*冬	风	VS	寒	风	冬	is semantically related	
	dōng	fēng		hán	fēng	to	寒.	
	winter	wind		cold	wind			
	cold wind							
undefined category	*应	尽	VS	应	该	尽	is related to 应 and	
	yīng	jǐn		yīng	gāi	该,	expressing the notion	
	should	fulfill		should	ought	of obligation.		
	should							

Note: The words with an asterisk are neologisms, while those without an asterisk are their idiomatic counterparts.

Critical reevaluation of Wang’s taxonomy reveals cross-category overlaps among several original classes (Wang, 2018). Some items originally assigned to “substitution of relevant meaning” or the “undefined category” are better understood under alternative structural labels. Accordingly, the present study consolidates the original 6 classes into 4 mutually exclusive neologism types: morphological substitution, blended misuse between word and morpheme, clipping-derived non-existents, and existent-structure non-existents, eliminating taxonomic ambiguity.

The broader literature distinguishes neology (the process of lexical innovation), neologisms (its tangible outputs, novel in form, meaning, or both; Varo et al., 2009; Cabré, 2015), and neonyms (a subset designating genuinely new concepts; Guerrero Ramos, 1995). While neologisms are typically intentional, they can also arise unintentionally, as in L2 errors that systematically apply morphological or phonological rules (Boulanger, 2010; Cabré, 2002; Suárez de la Torre et al., 2019).

This study adopts a broad definition of linguistic innovations novel in form or meaning (Varo et al., 2009; Cabré, 2015). It treats L2-induced nonce forms as psycholinguistically relevant neologisms. Prior work has theorized the learner-internal sources of these forms, including first language (L1) transfer, incomplete morphological representations, and overgeneralized word formation rules (Jiang, 2000, 2007; Xu, 2004; Wang, 2018; Everson, 1986; Hsu & Huang, 2000; Gao, 2004; Flege & MacKay, 2004; Yu, 2004). However, these accounts focus exclusively on production by L2 learners.

Existing scholarship on learner-generated neologisms thus centers on describing L2 output, but does not deploy these forms to test general theories of native morphological comprehension. A further methodological limitation is that corpus analyses and offline acceptability ratings alone cannot unpack the real-time inferential dynamics through which readers reconcile morphological anomaly with semantic interpretability. These gaps carry broad theoretical significance: competing models of lexical processing generate distinct predictions for exactly this class of non-canonical but interpretable input. To adjudicate between these competing accounts, this study combines an online self-paced reading task, which tracks incremental processing cost, with a supplementary offline acceptability judgment task, to uncover the computational principles governing native speakers' morphological inference.

2. Previous Research

2.1 Neological and pseudo-word processing

Recent empirical investigations into neological processing, though limited in scope, have begun to characterize the cognitive demands imposed by novel lexical forms. Suaza Restrepo et al. (2023) demonstrated that both neonyms and neologisms in Spanish required greater processing effort than established vocabulary in a visual lexical decision task. Yet, high general reading proficiency conferred no advantage for novel terms, suggesting that neological processing engages mechanisms distinct from those supporting familiar word recognition. Domain-specific knowledge, rather than reading ability per se, emerged as the critical facilitator for neonym recognition, with regular media exposure reducing these demands; notably, self-reported familiarity correlated only weakly with

actual recognition accuracy, implying that concrete experiential engagement outweighs subjective assessments. Wolfer and Klosa-Kückelhaus (2023) complemented this picture by showing, through mouse-tracking measures of response uncertainty, that English-derived neologisms in German were accepted more readily and with less hesitation than native German coinages, particularly among post-1980 cohorts. This generational gradient points toward an intriguing but underexplored possibility. The cognitive cost of neologism processing is modulated by the statistical alignment between the novel form and readers' prior linguistic exposure, including the ambient prevalence of borrowed lexical patterns.

At the neurocognitive level, Lamb (1999) and Pulvermüller (2002, 2005) have framed neological processing within associative network and functional web architectures, wherein contextual factors, frequency effects, and neural connectivity jointly shape novel word recognition. Brouwer et al. (2017) computationally modeled how cortical structures simulate word processing, suggesting that lexical innovation emerges from dynamic interactions between usage contexts and experiential factors. Varo (2020) extended this perspective by characterizing neological creation as an expression of cognitive flexibility, the capacity to generate novel interpretations from sensory and linguistic input. Yet, these theoretical accounts remain largely descriptive, offering broad architectural commitments without generating specific, testable predictions about how different types of structural deviations (e.g. substitution versus clipping) might differentially engage the proposed mechanisms. This theoretical underdetermination leaves unclear whether the observed processing patterns reflect general novelty detection or type-specific morphological computations.

Given the scarcity of research directly addressing Chinese neological processing, studies on pseudo-word recognition provide a useful but inherently limited proxy. Kapnoula et al. (2015) demonstrated that pseudo-words can achieve rapid lexicalization when embedded among familiar vocabulary, suggesting that the lexical system is highly sensitive to contextual statistics. EEG evidence (Di Liberto et al., 2019; McMurray et al., 2022) indicates that word versus pseudo-word differentiation occurs relatively late in processing, at disambiguation points or after multiple phonemes have been integrated, implying delayed neural discrimination. However, these studies have predominantly used pseudo-words that are phonotactically legal but morphologically unstructured, a poor approximation of Chinese neologisms, which systematically manipulate

existing morphemes within licensed or semi-licensed combinatorial templates. Brodbeck et al. (2018) and Gwilliams et al. (2018) further quantified surprisal effects at phonemic and lexical levels, showing heightened sensitivity to word versus pseudo-word sequencing, yet McMurray et al. (2022) found that highly regular pseudo-words in isolation elicited diminishing surprisal effects over time, suggesting that established lexical representations may not fully recruit higher-order processing networks without supportive context.

Individual and developmental differences further complicate the pseudo-word literature. While Stern et al. (2003) found that real-word stems similarly hindered both younger and older adults in rejecting Spanish pseudo-words, Lázaro et al. (2022) demonstrated that greater cognitive reserves afforded better resistance to morphological interference. Krishnamurthy et al. (2019) highlighted the role of grapheme-phoneme conversion in pseudo-word recognition, where visual input directly activates phonetic representations without semantic mediation. Developmental evidence (Szewczyk et al., 2018) reveals length effects and morpheme awareness difficulties in children, underscoring the protracted development of these mechanisms. However, these findings are largely language-specific and predominantly drawn from Indo-European languages, leaving open the question of whether analogous principles obtain in Chinese, an analytic language where the morpheme-syllable alignment and the absence of inflectional morphology may fundamentally alter the balance between compositional and holistic processing strategies when encountering novel morphologically complex forms. This cross-linguistic gap motivates the study focus on native Chinese speakers, for whom the processing of neologisms (systematically constructed from familiar morphemes but combined in non-canonical ways) offers a uniquely informative window into the interplay between structural deviance and contextual resolution.

2.2 Competing accounts of morphological processing

Debates over complex word comprehension have long revolved around how readers balance stored lexical knowledge against online structural analysis when encountering morphologically deviant but interpretable input. Many psycholinguistic frameworks fall under a broad family of morphology-sensitive processing accounts, which predict that processing difficulty will shift systematically according to the structural properties of the encountered word form, rather than

treating all surface anomalies as functionally equivalent. The augmented addressed-morphology model assumes parallel whole-word and morpheme-driven access routes: familiar established words are primarily retrieved as holistic entries, while unfamiliar or non-canonical forms trigger morphological decomposition into constituent morphemic units (Caramazza et al., 1988; Regel et al., 2019). Under this view, the magnitude of processing cost tracks the severity of morphological rule violation, so forms that break word-morpheme boundaries should produce greater reading time disruption than forms with milder substitution-based deviance. The dual route cascaded model and the dual mechanism model further elaborate such hybrid architectures. The dual route cascaded model posits concurrent whole-word lookup and morphological parsing pathways, with route weight modulated by lexical frequency and surface transparency (Clahsen, 1999). The dual mechanism model distinguishes rule-governed combinatorial decomposition for regular morphological patterns from memory-based retrieval for idiosyncratic, irregular forms, still expecting graded processing penalties tied to the nature of morphological violation (Clahsen, 1999).

Even single system associative frameworks that reject explicit symbolic morpheme rules also fall within this morphology-driven prediction family. The connectionist distributed representation morphological model implements learning through weighted statistical associations between orthographic form and semantic output, without hard-coded morpheme symbols, yet still generates graded processing difficulty conditioned on how well the surface structure of a stimulus aligns with learned statistical patterns (Rumelhart & McClelland, 1986; see Clahsen, 1999, for critical review). Similarly, the naive discriminative learning model, a general statistical learning architecture not purpose-built for morphology, derives morphological effects from learned co-occurrence relationships between surface cues and semantic outcomes. Though it contains no dedicated morphological parsing module, it still anticipates variable processing costs across structurally different deviant word forms, because distinct neological types supply different sets of predictive surface cues (Baayen, 2011). Taken together, these diverse frameworks, despite major architectural differences, converge on a shared empirical expectation: different categories of morphologically anomalous stimuli should produce measurable differences in real-time processing load.

This broad morphology-sensitive prediction family stands in sharp contrast to noisy-channel Bayesian decoding accounts of comprehension. Rather than prioritizing pre-specified

morphological structure analysis, noisy-channel models treat linguistic input as a potentially corrupted signal. Readers probabilistically infer the most plausible intended meaning, down-weighting surface form details when contextual evidence is sufficiently strong (Gibson et al., 2013; Cutter et al., 2024). The contrast between graded morphological-driven costs (uniting augmented addressed-morphology, dual route cascaded, dual mechanism, connectionist distributed representation morphological model, and naive discriminative learning model) versus uniform anomaly costs from noisy-channel Bayesian decoding forms the empirical test for the current study.

3. The Present Study

Driven by the research gaps outlined, a theoretically consequential question remained unanswered: how do native speakers, with their fully entrenched lexical representations, resolve the tension between morphological anomaly and semantic interpretability when encountering non-canonical forms? The present study investigated whether native Chinese speakers exhibit differential processing costs across 4 structurally distinct neological types (morphological substitution, blended misuse between word and morpheme, clipping-derived non-existent words, and existent-structure but non-existent words), and if so, what this reveals about the computational principles governing morphological processing.

Under the broader morphology-sensitive family of processing accounts, exemplified by compositional parsing approaches (Stevens & Plaut, 2022) that privilege morpheme-level analysis and assume that processing difficulty scales with the degree of structural deviance, neologisms violating word-integrity constraints such as blended misuse between word and morpheme should elicit longer reading times than those involving within-category morpheme substitutions such as morphological substitution. Such a pattern would support the view that the L1 system is sensitive to graded morphological violations, a prediction shared across multiple hybrid and statistical morphological frameworks reviewed above. For illustration, under this family of accounts, blended-misuse neologisms such as * 中学校 /zhōng xué xiào/ (mis-split from 中学 /zhōng xué/ ‘middle school’) should incur greater processing cost than morphological substitution items such as * 争场 /zhēng chǎng/ ‘battlefield’ (争 /zhēng/ replacing the morpheme 战 /zhàn/).

Alternatively, if native comprehension is better characterized by predictive, context-driven processing, whereby readers treat the input stream as potentially corrupted and infer the most probable intended message through Bayesian decoding over noisy evidence, then no significant reading-time differences should emerge across 4 neological types. In this scenario, the processor would treat all surface anomalies as equipotential deviations from canonical forms and resolve them through top-down contextual integration rather than type-specific morphological scrutiny. This null pattern would align with resource-rational principles of cognitive control, wherein the system adaptively re-weights the diagnosticity of morphological information against contextual coherence to optimize comprehension under uncertainty (Lieder & Griffiths, 2020; Ritz et al., 2022). Such an outcome would imply a sophisticated inference strategy that treats the input itself as inherently unreliable, a core commitment of noisy-channel models of language understanding (Gibson et al., 2013; Cutter et al., 2024). By establishing which of these competing architectures governs L1 neological resolution, this study extends the theoretical scope of neologism research to the study of cognitive adaptation to linguistic novelty.

4. Methodology

4.1 Participants

The study recruited 90 Mandarin Chinese speakers (mean age=20.15, SD=1.03, range=18-22) after obtaining informed consent. All participants had completed secondary education. The participants had received their entire education within the standard school system in China, ensuring adequate lexical knowledge for the experimental task. I excluded participants reporting cognitive or visual impairments. Based upon comprehension question accuracy following critical sentences, I excluded data from 9 participants, retaining 81 for analysis. The study received ethics approval from the institutional review board.

4.2 Procedure

In experiment 1, the participants completed a self-paced reading task on PCIBex Farm

(<https://farm.pcibex.net/>), a validated platform for psycholinguistic research (Frazier & Clifton, 1998). The materials comprised 40 critical sentences (containing neologisms) and 40 baseline sentences (without neologisms), with critical sentences evenly distributed across 4 neological types (10 sentences per type).

A priori power simulation for linear mixed-effects models was performed, simulating self-paced-reading-style reaction time data matching the design: 81 participants, 10 items per condition. The simulation yielded 84% statistical power to detect medium-sized condition differences of Cohen's $d=0.45$. Recognizing higher measurement noise in online PCIBex testing, I supplemented frequentist tests with Bayesian repeated measures ANOVA, which quantifies positive evidence for the null hypothesis instead of only relying on significance cutoffs, offsetting concerns about trial count per condition per participant.

Before the main task, the participants completed 5 practice trials matching the experimental sentences in length and structure. During the task, sentences appeared as sequences of masked words revealed sequentially via spacebar presses, implementing a non-cumulative moving-window paradigm (Fernández-Silva & Núñez Cortés, 2024; Mifka-Profozic et al., 2024; Shi et al., 2023). The system recorded response times for each keypress.

A separate group of 80 native Mandarin Chinese speakers (mean age=20.42, SD=1.21, range=18-23) participated in an offline acceptability judgment task as experiment 2. The materials consisted of the same stimuli distribution used in experiment 1. Each sentence was presented in its entirety by using PCIBex Farm as well. The participants rated each sentence on a 7-point Likert scale (1=completely unacceptable, 7=completely acceptable) based upon its naturalness and interpretability in modern Chinese. While the rest of the procedure resembled experiment 1, the participants had no time limit to offer a response. Hence, the participants' final, integrated grammatical judgment after full consideration of the stimuli was assessed, in contrast to experiment 1, which measured incremental processing cost.

4.3 Materials

Experimental stimuli were adopted from Wang's (2018) taxonomy of 6 neological types, retaining 4 mutually exclusive categories after merging overlapping classifications: (1) morphological substitution, (2) blended misuse between word and morpheme, (3) forming non-existent words by clipping, and (4) existent structure but non-existent word. 4 native Chinese research assistants verified the revised categorization scheme.

To isolate the processing cost attributable to lexical novelty per se, I constructed a matched baseline condition. For each critical sentence containing a neologism, I created a parallel baseline sentence by replacing the neologism with its conventional, idiomatic counterpart (e.g. * 争场 /zhēng chǎng/→战场 /zhàn chǎng/ 'battlefield'). All other linguistic elements remained identical, ensuring that critical and baseline sentences were matched precisely for syntactic structure, sentence length (in characters and words), position of the critical region, lexical content outside the critical region, overall semantic plausibility, and context. Furthermore, the idiomatic counterparts were selected to be high-frequency, concrete nouns (based upon the norming study), minimizing baseline retrieval difficulty. This design allows us to attribute any significant slowdown in the critical conditions specifically to the encounter with a novel lexical form, rather than to general sentence complexity or conceptual unfamiliarity.

To ensure proper counterbalancing, baseline sentences were distributed across the same 4 conditions using the Latin square design applied to critical sentences. Each task version included baseline sentences corresponding to all 4 neological conditions, preventing sentence repetition within a single version. This design ensured that participants were exposed to both critical and baseline sentences across all conditions in different task versions, thereby mitigating potential order effects and maintaining comparability of processing times. For illustration, **Tables 2** to **Table 5** present example sentences from each condition, including glossing. The neologisms in the critical regions are highlighted in red, while their idiomatic counterparts (not shown to participants) appear in brackets. The word following the critical region belongs to the post-critical region. All sentences adhered to the structure: "NP₁+VP₁+NP (neologism) genitive+NP₃, and VP₂+NP₄".

Table 2. Condition 1: morphological substitution

导演	叙述了	争场（战场）的	故事情节，
dǎo yǎnhái	xù shù le	zhēng chǎng de	gù shì qíng jié
director	narrate-PST	battlefield GEN	storyline
还	介绍了	演员	
hái	jiè shào le	yǎn yuán	
and	introduce-PST	actor	
‘The director narrated the storyline of the battlefield and introduced the actors.’			

(Sample comprehension question: 是家长介绍了演员吗？‘Did the parents introduce the actors?’)

Table 3. Condition 2: blended misuse between word and morpheme

学生	学习了	中学校（中学）的	基础课程，
xué shēng	xué xí le	zhōng xué xiào de	jī chǔ kè chéng
student	study-PST	secondary school GEN	basic course
还	获得了	证书。	
hái	huò dé le	zhèng shū	
and	obtain-PST	certificate	
‘Students studied the basic courses of the secondary school and obtained certificates.’			

Table 4. Condition 3: forming a non-existent word by clipping

教授	讨论了	独孩（独生子女）的	思维特点，
jiào shòu	tǎo lùn le	dú hái de	sī wéi tè diǎn
professor	discuss-PST	only child GEN	thinking style
还	报告了	数据。	
hái	bào gào le	shù jù	
and	report-PST	data	
The professor discussed the characteristics of an only child’s thinking style and reported data.			

Table 5. Condition 4: existent structure but non-existent word

工人	保障了	财主人（财主）的	生命安全，
gōng rén	bǎo zhàng le	cái zhǔ rén de	shēng mìng ān quán
worker	safeguard-PST	rich man GEN	life safety
还	关心了	老人。	

hái	guān xīn le	lǎo rén
and	care-PST	the old

The worker safeguarded the rich man's life safety and cared about the old.

The experimental materials underwent rigorous validation. First, 30 native Chinese speakers rated the concreteness of 66 nouns from Wang's (2018) study on a 5-point Likert scale (1=abstract, 5=concrete). 40 nouns with ratings between 4.0 and 5.0 ($SD=0.36$) were selected as neological sources. Additionally, the frequency of individual characters was cross-checked against the Center for Chinese Linguistics PKU (CCL) corpus to ensure all characters had a frequency below 3000. This upper-bound control was implemented to prevent any single character from being exceptionally low-frequency (and thus artificially difficult to retrieve). Processing differences across conditions should be more likely attributable to the experimental manipulation of lexical novelty rather than to extreme variations in character familiarity. 40 neologisms were verified as either non-existent or rarely used in a modern Chinese corpus (i.e. CCL). Although stroke number and morphological structure were not strictly controlled, prior research indicates their impact is negligible when items are closely matched on frequency and semantic characteristics (Diependaele et al., 2013; Lemhöfer et al., 2008). **Moreover, minor length/stroke variation is intrinsic to naturally occurring L2-derived neologisms. Rigid matching would distort their genuine structural properties and harm ecological validity. The subsequent reporting that online reading yielded no reliable reading time differences across neologism types demonstrates that minor lexical variables exerted limited influence. Otherwise, systematic condition-level processing differences should have emerged.** The primary focus of this study was the experimental manipulation of lexical novelty, which was expected to outweigh minor structural effects.

10 native Chinese speakers with at least an undergraduate degree evaluated the acceptability of the critical sentences after their neologisms had been replaced with the corresponding idiomatic forms, as well as the baseline sentences. This criterion was applied to ensure that evaluators had sufficient metalinguistic awareness and exposure to formal written Chinese to make consistent, reliable judgments about sentence acceptability, independent of dialectal or informal usage influences. They rated all sentences by using a 5-point scale (1=completely unacceptable, 5=completely acceptable). The mean acceptability rating was 4.88 ($SD=0.13$). Finally, 6 additional native speakers reviewed

the materials for linguistic accuracy and provided feedback on word choice and sentence construction before the formal experiment.

5. Results

5.1 Experiment 1: self-paced reading task

The participants were instructed to read the sentences at their own pace while maintaining comprehension. A follow-up question assessed their understanding and attentiveness during sentence comprehension (mean accuracy rate=87%, SD=3.67%). The participants with accuracy rates below 80% were excluded from the analysis, resulting in a 10% data loss, likely due to inattentiveness or poor comprehension. The entire experiment lasted approximately 25 minutes.

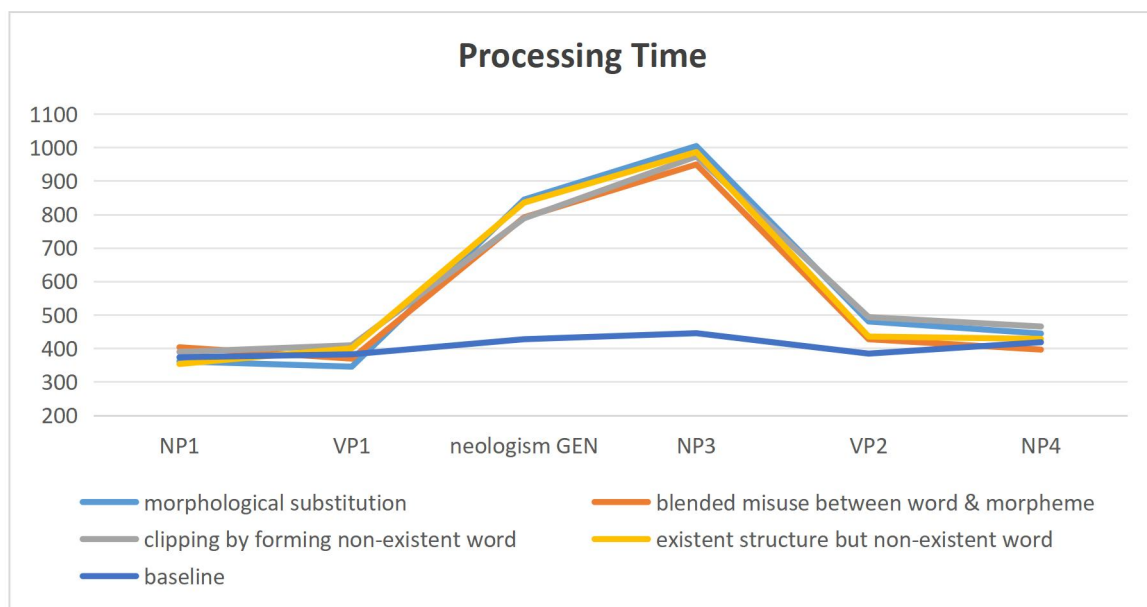
To examine the statistical significance of processing times across 4 neological types in the critical region, descriptive statistics were calculated. The independent variables were 4 neological types and the baseline condition, with processing time serving as the dependent variable. The results revealed the following average processing times (in milliseconds): morphological substitution elicited the longest processing time (mean=844ms), followed by existent structure but non-existent word (mean=835ms), blended misuse between word and morpheme (mean=791ms), and forming a non-existent word by clipping (mean=788ms), which showed the shortest processing time among the neological conditions. In contrast, the baseline condition (featuring idiomatic expressions) demonstrated a substantially shorter average processing time (mean=427ms). These descriptive results suggest that morphological substitution imposed the greatest processing load, while clipped neologisms were processed with relatively less difficulty.

To determine whether these differences were statistically significant, post hoc multiple comparisons using ANOVA were conducted to examine the relationships among 4 neological types and the baseline condition (see also Gibson et al., 2017; Myors & Murphy, 2023; Sonderegger, 2023). Due to violations of the homogeneity of variance and normality assumptions, the Games-Howell test was employed (see also Lim & Godfroid, 2015). The inferential results indicated no statistically significant differences between any pair of neological types. For instance, the comparison between

morphological substitution and existent structure but non-existent word yielded a mean difference of 9ms, 95% CI [-127, 145], $p=0.98$; between morphological substitution and clipping, mean difference=56 ms, 95% CI [-55, 167], $p=0.32$; between blended misuse and clipping, mean difference=3ms, 95% CI [-108, 114], $p=0.99$. In contrast, each neological condition differed significantly from the baseline condition, with all $p<0.001$. These findings confirm that while neologisms collectively imposed greater processing costs than idiomatic words, 4 neological types did not differ meaningfully in processing difficulty within the critical region.

The post-critical region, which often reflects delayed integration effects, was also analyzed. Descriptive statistics indicated that average processing times in this region were generally longer than those in the critical region (see **Figure 1**). Morphological substitution again showed the longest processing time (mean=1004ms), followed by existent structure but non-existent word (mean=986ms), clipping (mean=973ms), and blended misuse (mean=949ms). The baseline condition, by comparison, required significantly less time (mean=445ms), maintaining its advantage in processing efficiency for idiomatic input.

Figure 1. Processing timing in a line chart



As with the critical region, these descriptive results were analyzed by using Games-Howell post hoc comparisons. Once again, no statistically significant differences emerged between any two neological types. The p -values ranged from 0.11 (morphological substitution versus existent structure but non-existent word) to 0.96 (morphological substitution versus clipping), with all

comparisons failing to reach significance at $p < 0.05$. In contrast, all neological conditions differed significantly from the baseline condition ($p < 0.001$), confirming that neologisms consistently demanded greater processing effort, even beyond the critical word boundary.

To account for potential variability among participants and items, I applied a linear mixed-effects model with random effects (Koutamanis et al., 2024; Martínez-Huertas et al., 2022). The model specification was $\log(\text{time}) \sim \text{condition} + (1|\text{participant}) + (1|\text{item})$. The descriptive and ANOVA-based comparisons reported above were conducted on the full dataset without outlier removal, as the non-parametric Games-Howell test is robust to outliers and its use aligns with the violated assumption of homogeneity of variance. For the subsequent linear mixed-effects modeling, which assumes normally distributed residuals, I applied a more conservative preprocessing step: response times exceeding 3 standard deviations from the condition mean were removed (affecting $< 2\%$ of trials) (Baayen & Milin, 2010). This two-step approach ensures that both robust non-parametric comparisons and sensitive parametric model estimates are reported, maximizing analytical rigor. The mixed model revealed that when including participants and items as random effects, the main effects of neological conditions remained nonsignificant in both the critical ($\chi^2(3) = 1.96$, $p = 0.582$) and post-critical ($\chi^2(3) = 1.78$, $p = 0.474$) regions. However, significant differences emerged between neological and baseline conditions in both the critical ($p = 0.02$) and post-critical ($p = 0.03$) regions, confirming distinct perceptual differences between neologisms and idiomatic items. The inclusion of random intercepts accounted for variability across participants and items while maintaining methodological rigor. The analysis opted not to include random slopes to avoid model overfitting and maintain stability, given the sufficient statistical control achieved. **Pairwise contrasts from the linear mixed-effects models further supplied coefficient estimates and 95% confidence intervals, all of which overlapped with zero for neologism-to-neologism comparisons.**

Given that conventional null hypothesis significance testing cannot provide evidence for the absence of an effect, I additionally conducted Bayesian repeated measures ANOVA on the critical region reading times to quantify support for the null hypothesis across 4 neological types. Bayesian analysis provided positive support for the null hypothesis across all pairwise comparisons (all $\text{BF}_{01} > 3.5$), indicating that the absence of significant differences is not merely due to insufficient statistical power. Crucially, when restricting analysis to the participants with $> 90\%$ comprehension

accuracy (thereby minimizing potential shallow engagement), the null pattern remained unchanged, arguing against the concern that deviant stimuli induced superficial processing.

5.2 Experiment 2: grammaticality judgment task

The independent variables were 4 neological types and the baseline condition, with acceptability rating serving as the dependent variable. The results revealed the following mean ratings (on a 7-point scale): morphological substitution elicited the lowest acceptability (mean=2.93, SD=0.47), followed by existent structure but non-existent word (mean=2.95, SD=0.51), blended misuse between word and morpheme (mean=3.01, SD=0.49), and forming a non-existent word by clipping (mean=3.15, SD=0.46), which showed the highest acceptability among the neological conditions. In contrast, the baseline condition (featuring idiomatic expressions) demonstrated a substantially higher mean rating (mean=6.54, SD=0.38). It appears that morphological substitution was perceived as least acceptable, while clipped neologisms were judged relatively more acceptable than other neological types.

To determine whether these differences were statistically significant, a one-way ANOVA was conducted to examine the relationships among 4 neological types and the baseline condition. A one-way ANOVA revealed no significant main effect of neological type ($p=0.24$). Due to violations of homogeneity of variance and normality assumptions, the Games-Howell test was employed, consistent with the analytical approach used in experiment 1. The inferential results indicated that the difference between clipping and morphological substitution approached significance (mean difference=0.22, 95% CI [0.002, 0.438], $p=0.048$), while comparisons among other neological pairs remained non-significant. For instance, morphological substitution versus existent structure but non-existent word yielded a mean difference of 0.02, 95% CI [-0.20, 0.24], $p=0.91$. In contrast, each neological condition differed significantly from the baseline condition, with all $p<0.001$. These findings suggest that clipped neologisms were rated as marginally more acceptable than other neological types, though neologisms collectively were judged significantly less acceptable than idiomatic words.

To account for potential variability among participants and items, a linear mixed-effects model was

applied with random intercepts for participants and items. The model specification was $\text{rating} \sim \text{condition} + (1|\text{participant}) + (1|\text{item})$. The mixed model revealed that when including participants and items as random effects, the main effects of neological conditions remained nonsignificant ($\chi^2(3) = 7.14, p=0.25$), with pairwise contrast between clipping and morphological substitution reaching significance ($p=0.045$). Significant differences emerged between neological and baseline conditions ($\chi^2(3)=4.32, p<0.01$). The inclusion of random intercepts accounted for variability across participants and items while maintaining methodological rigor. As in experiment 1, random slopes were not included to avoid model overfitting and maintain stability. Pairwise contrasts from this model returned coefficient estimates and 95% CIs; only the contrast between clipping-derived non-existent words and morphological substitution excluded zero.

To move beyond the limitations of null hypothesis significance testing, I again conducted Bayesian analysis on the rating data. Bayesian analysis provided positive support for the null hypothesis among the three lower-rated neological types (all $\text{BF}_{01} > 3.5$), but showed anecdotal evidence for a difference between clipping and morphological substitution ($\text{BF}_{10} = 2.1$). Hence, the observed advantage for clipped neologisms is not merely due to insufficient statistical power.

6. Discussion

The absence of differential processing costs across typologically distinct neologisms challenges the intuitive assumption that structural deviance scales linearly with comprehension difficulty. Instead, the findings invite a reconceptualization of the L1 lexical system as a rational inference engine that dynamically calibrates its sensitivity to morphological information based upon the statistical structure of the task environment. This perspective draws upon two complementary frameworks, namely noisy-channel Bayesian inference and adaptive resource allocation, which together offer a unified account of how entrenched linguistic systems maintain stability when confronted with non-canonical input.

From the noisy-channel perspective, language comprehension is fundamentally a problem of recovering intended messages from potentially corrupted signals (Cutter et al., 2024; Gibson et al.,

2017; Ryskin et al., 2018; Ryskin et al., 2021; Zhang et al., 2023). The comprehender does not simply accept the input at face value but rather computes the posterior probability of candidate intended sentences given the observed surface form, integrating prior expectations about plausible meanings with a likelihood model of how transmissions might be distorted. Critically, the likelihood term in this Bayesian decoding framework is not indifferent to the nature of the corruption: string edits involving deletions are assigned higher probability than insertions due to the Bayesian size principle, and the overall noise rate of the communicative situation is inferred from context (Gibson et al., 2013). With reference to current data, the participants encountering a neologism such as 火站 /huǒ zhàn/ ‘train station’ did not engage in a morphological decomposition process that would be differentially sensitive to whether the error arose from substitution, clipping, or boundary misanalysis. Rather, the processor treats the observed form as a noisy realization of some intended lexical item and computes, over the space of possible intended words, the posterior distribution conditioned upon the perceptual evidence and the prior plausibility of candidate meanings in the current context. If different neological types (despite their distinct generative mechanisms) produce surface forms that are roughly equally distant from their most plausible intended targets under a plausible noise model, then the resulting posterior probabilities will be similar, yielding comparable processing times.

This account resonates with recent evidence that comprehenders rationally integrate noisy evidence with prior semantic expectations even when the noise affects syntactic function words (Cutter et al., 2024), and that such integration is modulated by the perceived base rate of errors in the communicative environment. The likelihood of a given neologism being generated from its idiomatic counterpart depends upon the edit distance between two forms. In the present materials, 4 neological types may not have differed substantially in their edit distances to the intended lexical items once the possibility of phonological or orthographic proximity is considered. For instance, morphological substitution and word-morpheme misuse both involve changes that could be characterized as single-morpheme substitutions or deletions under a character-level noise model. The observed uniformity in processing times across types is therefore precisely what a noisy-channel model would predict if the likelihood term $P(s_i \rightarrow s_p)$ is comparable across conditions.

However, the noisy-channel account alone does not fully explain why native readers would even

entertain the possibility that a well-formed, familiar-looking form might be a corruption rather than a genuine lexical item. This is where the adaptive encoding perspective offers a key complement. Under the framework of resource-rational analysis (Lieder & Griffiths, 2020), cognitive systems should adaptively allocate processing resources to the dimensions of input that are most diagnostic for achieving task goals while de-emphasizing less informative dimensions. In the online self-paced reading task, the primary goal is to maintain comprehension for subsequent accuracy questions, not to perform explicit grammaticality judgments or to identify morphological violations. Once a reader implicitly infers odd but still interpretable words, an optimal strategy is to down-weight the diagnosticity of morphological well-formedness and up-weight contextual coherence, a form of control allocation that is sensitive to the expected utility of different processing policies (Shenhav et al., 2013). The offline acceptability judgment task, by contrast, taps into a different processing regime in which readers have the opportunity for reflective evaluation, potentially revealing sensitivity to qualitative differences among deviation types that are suppressed during online reading.

The offline data bore out this prediction. Clipping and morphological substitution represented the two extremes in online processing times, fastest versus slowest, and the offline acceptability data aligned with this asymmetry. Clipping-derived neologisms received marginally higher acceptability ratings than the other three types, a difference that was absent in the online reading times. This pattern suggests that clipping, as in 独孩 /dú hái/ from 独生子女 /dú shēng zǐ nǚ/ ‘only child’, represents an overapplication of a legitimate word-formation rule, making it structurally transparent and compositionally recoverable. Morphological substitution (内边 /nèi bian/ for 里边 /lǐ bian/ ‘inside’), by contrast, involves the incorrect selection of a synonymous but contextually inappropriate morpheme, which triggers a different kind of metalinguistic judgment even though it imposes no greater online burden. The offline task thus taps into a post-comprehension evaluation stage at which the cognitive system differentiates between rule extension and rule misapplication, a distinction that does not surface during the temporally constrained demands of real-time reading (Ferreira & Patson, 2007; Ferreira et al., 2002). This finding enriches the noisy-channel framework by revealing that the likelihood term is not monolithic: different types of deviations, even when they impose comparable online costs, can be assigned different likelihood values when sufficient time is available for evaluation.

The adaptive re-weighting can be understood as the cognitive system solving a multivariate optimization problem in which multiple control signals (including attentional enhancement of target-relevant information and suppression of task-irrelevant dimensions) are configured to maximize reward rate (Ritz et al., 2022; Leng et al., 2021). In the present context, the target is the intended message, while distractor information includes the precise morphological composition of the critical word. By preferentially processing contextual and thematic information, readers effectively regulate the influence of morphological irregularities on their incremental interpretation. This strategic reconfiguration aligns with the broader principle that cognitive control is deployed not uniformly but rather in a manner that reflects the expected costs and benefits of alternative control configurations (Shenhav et al., 2017).

Our predictive processing machinery continually learns the predictive validity of different cues and adjusts its weighting accordingly (Gong & Wei, 2025). This aligns with the broader notion that language processing operates under resource-rational constraints, optimizing the use of limited computational resources to achieve reliable comprehension (Lieder & Griffiths, 2020). The present findings suggest that the L1 lexical system rapidly learns that morphological deviance has low predictive value for task success, thereby compressing the apparent differences among neological types. This interpretation gains support from research on cognitive effort discounting, which shows that individuals are willing to forego rewards to avoid tasks requiring high cognitive demand, and that effort costs are best characterized as quadratic functions of the intensity of engagement (Vogel et al., 2020; Massar et al., 2020).

The convergence of the noisy-channel and adaptive-encoding frameworks points toward a more general principle: the language system treats incoming signals as inherently uncertain and adjusts its inference strategies to match the reliability of different information sources. This is a core feature of the cognitive architecture that supports robust communication in the face of ubiquitous noise, including dialectal variation, speech errors, environmental interference, and individual differences in production (Christiansen & Chater, 2016). The current stimuli, though artificially constructed from learner errors, tap into this fundamental adaptive capacity, revealing that native processing is characterized by a sophisticated interplay between prior linguistic knowledge, likelihood estimation

under uncertainty, and online tuning of control policies.

The reconceptualization has implications beyond the present experimental paradigm. If the language system is fundamentally a rational estimator over uncertain input, then phenomena traditionally interpreted as processing limitations, including garden-path effects, grammaticality illusions, and even some forms of syntactic processing difficulty, may be re-evaluated as manifestations of optimal inference under resource constraints (Futrell et al., 2020). Lossy-context surprisal, which posits that processing difficulty reflects surprisal given a memory representation that has undergone information loss, provides a formal framework for understanding how expectations and memory limitations interact (Futrell et al., 2020). The present findings extend this logic by demonstrating that native speakers treat the input stream itself as potentially lossy or corrupted, and that their processing difficulty is shaped not only by what information is lost from memory but also by what information is treated as unreliable in the current task environment.

Finally, several limitations of the current study warrant discussion. First, the current design implicitly assumed that native speakers uniformly recognize a given neologism as a deviation. However, some neologisms might be misperceived by certain participants as rare but existing lexical items rather than as clear errors. This issue may introduce unmeasured variance in the “deviance” variable itself, as processing times for identical stimuli may reflect either novelty detection or failed lexical retrieval, depending on individual lexical repertoires. Future studies could include a post-trial familiarity judgment (e.g. “Have you seen or used this word before?”) to identify and model individual differences in perceived lexical novelty as a covariate. Second, one of the neological categories, namely existent structure but non-existent words, instantiates statistical preemption, where valid compound templates produce candidates that never conventionalize. The other three categories involve real morphological and morpheme-boundary damage. While the design compares these two forms of deviance, learner-derived stimuli can’t fully disentangle processing costs stemming from preemption versus morphological corruption, a limitation I note alongside a suggestion for future artificial stimulus experiments.

7. Conclusion

By treating L2-learner-generated neologisms as probes into the architecture of native morphological processing, the present study reveals that the L1 lexical system operates as a rational, adaptive inference engine rather than a rigid morphological parser. Crucially, it goes beyond merely extending noisy-channel theory to a new stimulus domain. The dissociation between online reading times, which compressed all neological types into a uniform processing cost, and offline acceptability judgments, which selectively favored clipping-derived forms, demonstrates that the likelihood term in Bayesian decoding is not a fixed property but dynamically re-weights across processing regimes. A finer-grained architecture is observed in which rapid comprehension proceeds via heuristic inference, while reflective evaluation recruits a more differentiated assessment of deviation types. The lexical system thus exemplifies a general cognitive principle, namely adaptive stability amid variability, which underscores the deep continuity between linguistic processing and the broader architecture of cognition.

Disclosure Statement

Conflict of interest: There is no conflict of interest.

Ethics approval: This study received ethical clearance from Survey and Behavioural Research Ethics Committee (SBREC), The Chinese University of Hong Kong

Consent to participate: Informed consent was obtained from all individual participants in the study.

Data availability: https://osf.io/s97x3/?view_only=b6fbe26039834a98914965eb791d3805

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